Tancrède AMIEL

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Research interests: Mathematics for machine learning (optimization, probability, information theory) — AI for robotics (physical AI, RL)

EDUCATION

CENTRALESUPÉLEC — Plateau de Saclay, FR
Master of Engineering (Diplôme d'ingénieur), General Engineering — *2024 – 2028*
Entrepreneurship Track — Gap year 2026–27 (internship in Tokyo), Master's level — GPA: **3.94**/4.33 (Year 1), **4.03**/4.33 (Year 2)
Probability & Statistics: **Probability, Statistics, Machine Learning** — Optimization & Analysis: **Optimization, PDEs**
Information & Quantum: **Information Theory, Quantum Physics & Computing** — Computing: **Algorithms, HPC, Cloud**

COLLÈGE STANISLAS — Paris, FR
*Scientific Preparatory Classes (CPGE) — PCSI/PC** — *2022 – 2024*
Two-year intensive program in advanced mathematics and physics (PC*: advanced stream), preparing for national competitive exams

RESEARCH & ACADEMIC PROJECTS

HPC OPTIMIZATION — CentraleSupélec – CEA
Route optimization under environmental constraints — Grade: 19/20 — *Feb. – Apr. 2026*
Designed a pipeline computing optimal routes over realistic 3D terrain, minimizing a simulated physics-based risk cost

- HPC & Simulation: **ARMEN (CEA) and Δ-stepping algorithms, parallelized (MPI) on a supercomputer (TGCC)**
- Optimization & Algorithms: **Bayesian optimization (Gaussian process surrogate model), raycasting**

RESEARCH PAPER ANALYSIS – Variational Tensor Neural Networks — CentraleSupélec
Information Theory & Quantum Information course — written critical report (LaTeX) and oral presentation — *June 2026*

ACOUSTIC OPTIMIZATION — CentraleSupélec
Indoor noise pollution control through energy distribution optimization — *Nov. 2025*
Minimization of a concert hall's acoustic energy through optimal distribution of a porous material on a wall

- Mathematics: **PDE-constrained optimization (Helmholtz–Robin, fractal boundary), adjoint method, projected gradient**
- Simulation: **2D propagation of a musical chord in the hall, effect of the absorbing material on its energy distribution**

ENTREPRENEURSHIP TRACK – AutoGear — CentraleSupélec
Automatic bicycle derailleur designed to optimize rider effort — Grade: 17/20 — *Jan. 2025 – Present*

- Hardware: **Plug-and-play MVP, embedded electronics, Arduino, 3D printing**
- Software: **Supervised learning (XGBoost, RF, MLR, SVM), dataset building & cleaning, feature engineering**

FRENCH TOURNAMENT OF YOUNG MATHEMATICIANS — Animath – TFJM²
National team research competition on open problems — 3rd place, Paris tournament — *Jan. – May 2022*
Five months of research on a combinatorics problem, followed by presentation and defense of the results before a jury of researchers

- Problem: **Recovering an unknown seating-plan permutation with a minimum number of questions under constraints**
- Proofs: **Optimality via worst-case analysis, comparison of strategies, design and analysis of decision algorithms**

INDUSTRY EXPERIENCE

AI / ML ENGINEER – AI Solution Design & Deployment — Tokyo, JAPAN
Dassault Systèmes K.K. — *Sept. 2026 – Feb. 2027*
Driving internal adoption of generative AI, from business use cases to designing, implementing, and hosting solutions on open-weight models

- Development: **RAG pipelines, MCP tools connected to internal models**
- Strategy (Japanese subsidiary): **use-case prioritization (impact, complexity), hosting sizing and costing (servers, GPUs)**
- Second phase: **Physical AI for robotics (Reinforcement Learning, Computer Vision)**

DEPUTY PROJECT MANAGER (IAgriM) – SOLUTION ARCHITECT — Paris – Bamako, MALI
Danew Talla Electronics — *Jul. 2024 – Present*
Development of an interface powered by a local AI to provide Malian farmers with information offline

- Conceived and pitched the concept, approved by the Ministers of Digital Economy and of Employment; consortium with the Mali Center for AI and Robotics (CIAR) and the NGO AgriCap

TEACHING

ORAL EXAMINER (COLLEUR) IN MATHEMATICS – PC* AND PCSI — Collège Stanislas, Paris
Mathematics oral examinations (colles) in scientific preparatory classes — *June 2025 – June 2026*

PRIVATE TUTOR IN MATHEMATICS AND PHYSICS — Paris, FR
Middle/high school students; a PC* student through to his engineering school admission — *Sept. 2024 – June 2026*

VOLUNTEER WORK

CHEERUP! CENTRALESUPÉLEC — CentraleSupélec
Young cancer patients: hospital visits and outings; communication and art direction for fundraising events — *Oct. 2024 – Feb. 2026*

SKILLS

IT — Python (Pandas), LaTeX, OpenMP/MPI, SLURM, Linux CLI, Git, Docker, basics: C++/JS/CSS/HTML
Languages — French (native) — English (C1/C2, bilingual) — Spanish (B2, DELE certified)